# Ethical Issues in Research

# Agenda

- Right of People directly involved
- Responsibilities of an Ethical Researcher
- Data Privacy and Security
- Plagiarism and Intellectual Property
- Algorithmic Bias and Fairness
- Environmental Responsibility
- Conflict of Interest
- Copyright and Patent
- ACM Code of Ethics and IEEE Code of Ethics

- Ethics may be defined as:
  - Sets of moral principles or norms that are used to guide moral choices of behavior and relationships with others. (Blumberg, et al, 2005)
  - The moral principles and actions guiding and shaping research. (ESRC, 2004)

- Ethics in research - - The set of values, standards, and principles used to distinguish between appropriate and acceptable conduct and unacceptable conduct at all stages of the research process.

# Introduction to Research Ethics

- Research ethics = norms for acceptable/unacceptable behavior in research.
- Ensures integrity, quality, and accountability in research.
- Ethical principles guide research design, conduct, and publication.

- "Ethics is knowing the difference between what you have a right to do and what is right to do." – Justice Potter Stewart



# Ethical Considerations for *Participants* in Research (6Ps)

- **Participants**
  - **Direct Participants**: Treat individuals directly involved in your research (interviewees, observed subjects, questionnaire respondents, or document providers) with respect, fairness, and honesty.
  - **Researcher(s)**: As a researcher (or part of a team), act ethically, maintain integrity, and ensure transparency in all research processes.
  - **Academic Community**: Consider the impact of your research on readers, reviewers, and learners who engage with your work, ensuring accuracy and credibility.
  - **End-Users**: Be mindful of individuals who may use or be affected by any computer-based products or outcomes of your research, prioritizing their safety and well-being.
- Treat all participants, whether directly or indirectly involved, **fairly and with honesty**, adhering to ethical research standards.

# Ethical Clearance in Research

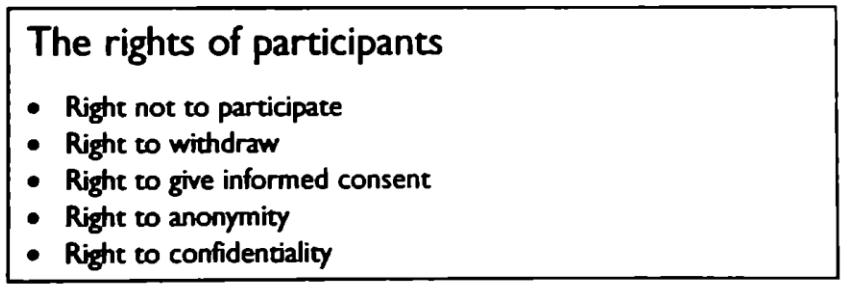
- **Ethical Consideration at Design Stage**: Institutions require ethical aspects to be addressed during the research design phase, ensuring responsible planning.
- **Ethical Clearance Mandate**: Researchers must obtain ethical approval before commencing any research project.
- **Clearance Procedures**: Processes vary by institution but typically involve detailing the project's purpose, funding, participant involvement, and ethical considerations.
- **External Scrutiny**: Research ethics committees, composed of experienced members, review the project to ensure ethical standards are met.
- **Core Ethical Principles**: Committees assess whether participants and affected individuals will be treated fairly, with dignity, and without harm.
- **Approval or Redesign**: *Projects are approved if ethical standards are satisfied*; otherwise, researchers must revise and resubmit for clearance.

# The Law and Research

- **Legal Compliance**: Ensure all research activities, from **literature review** to **final outcomes**, adhere to **national laws** and regulations.
- **Data Protection**: Understand **data protection laws**, including individuals' **rights** and researchers' **responsibilities** when handling **personal data**.
- **Participant Incentives**: Verify the **legality** of offering incentives (e.g., **prize draws**) to encourage participation in your research.
- **Intellectual Property**: Clarify **ownership rights** for **images**, **software**, **theses**, and **publications**, respecting **copyright** and **intellectual property laws**.
- **Technology Restrictions**: Be aware of **legal constraints** on technology use, such as **internet access**, **encryption software**, and sharing **innovations internationally**.
- **Legal Liability**: Consider the **legal responsibilities** of **software developers** for the systems they design and create.
- **Ethics vs. Legality**: Note that research can be **legal but unethical**; always prioritize **ethical considerations** alongside **legal compliance**.

# Rights of People Directly Involved in Research

- **Terminology Variation**: Participants are referred to differently based on research strategies:
  - **Experiments**: Research **subjects**
  - **Surveys**: Research **respondents**
  - **Case Studies/Ethnography**: Research **informants**, **members**, or **participants**
  - **Action Research**: **Participants** or **co-researchers** (in emancipatory forms).
- **Hierarchy of Involvement**: Terminology reflects the level of involvement, from **passive subjects** in experiments to **active co-researchers** in action research.
- **Human Rights**: All participants, regardless of their role, are **humans** with the right to be treated with **dignity** and to **benefit** from the research.
- **Avoid Exploitation**: Participants should not be **used** solely for the researcher's or funder's advantage; research must be both **ethical** and **legal**.
- **Protection from Harm**: Ensure participants face no **adverse consequences**—physiological, psychological, social, political, or economic.
- **Collectives' Rights**: Organizations, communities, or groups studied (e.g., businesses, hospitals, schools) have the same rights as individuals to be treated **fairly** and with **dignity**, and to avoid **ill-effects**.



# Rights of Participants

- **Right Not to Participate in Research**
  - **Voluntary Participation**: Individuals or organizations have the **right to decline** participation in research without facing pressure, coercion, or threats. Their decision must be respected.
  - **Power Dynamics**: Researchers in positions of authority (e.g., managers, teachers) must ensure participation is **truly voluntary** and not influenced by their power over subordinates or students.
- **Right to Withdraw from Research**
  - **Withdrawal at Any Time**: Participants have the **right to withdraw** from the research at any stage, even if it disrupts or compromises the study.
  - **Partial Participation**: Participants can **opt out of specific parts** of the research, such as declining to answer certain questions or refusing to participate in particular activities.

# Rights of Participants

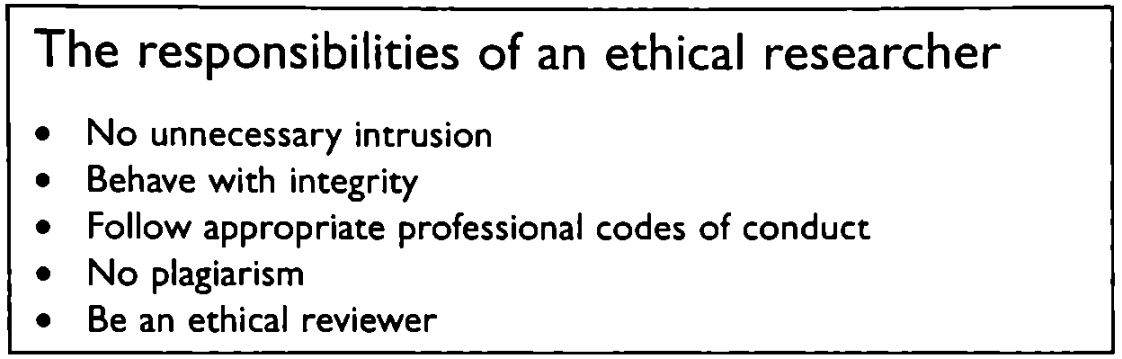
- **Right to Give Informed Consent**
  - Individuals, and collections of individuals, have the right to give informed consent.
  - **Full Disclosure**: Participants must be **fully informed** about the research purpose, their involvement, the researchers' details, the sponsoring organization, activities required, time commitment, incentives, and data usage.
  - **Voluntary Participation**: Participants must be informed of their **right to decline** or **withdraw** at any time, and consent should be **written** but not treated as a binding contract.
  - **Exceptions to Consent**: In cases where **informed consent** is not feasible (e.g., to avoid bias or in covert observation), researchers must justify this to the ethics committee, ensure no harm, and provide a **debriefing** afterward.
  - **Special Cases**: For participants unable to give consent (e.g., children or ill individuals), seek permission from **guardians** or **relatives**, but involve the participants as much as possible. Consent is an **ongoing process**, especially in long-term studies, and may require **renegotiation** and **renewal**.

# Rights of Participants

- **Right to Anonymity**
  - **Protecting Identity**: Participants have the right to have their **identity** and **location** protected, often through the use of **pseudonyms**, gender changes, or other disguises to prevent recognition.
  - **Organizational Anonymity**: Organizations' identities should also be **disguised** in research reports unless they explicitly grant permission, which may involve reviewing the report and discussing its dissemination.
- **Right to Confidentiality**
  - **Secure Data Handling**: Participants have the right to have their data kept **confidential** and secure. Researchers must ensure data is not left accessible to others and is stored safely.
  - **Respecting Confidential Requests**: If participants share information **"in confidence"**, it must not be included in the research report unless they explicitly release the researcher from this obligation. Similarly, organizations may require findings to remain confidential, necessitating prior discussions about **disguising identities** or agreeing on **delayed publication**.

# Responsibilities of an Ethical Researcher

- **Respect Participants' Rights**: Ensure **anonymity** and **confidentiality** for participants, and avoid **coercion** or misuse of power to secure participation.
- **Informed Consent**: Obtain **informed consent** from participants, avoiding deception unless absolutely necessary and no harm is caused.
- **Honesty and Integrity**: Conduct research with **honesty**, **transparency**, and **integrity**, ensuring accuracy and reliability in data collection and reporting.
- **Accountability to the Academic Community**: Share findings responsibly, contribute to the advancement of knowledge, and adhere to ethical standards expected by the **academic community**.



- **No Unnecessary Intrusion**
  - **Avoid Redundant Research**: Ensure the information you seek is not already available, and avoid intruding into participants' activities unnecessarily.
  - **Minimize Intrusive Questions**: Design research tools (e.g., questionnaires) to avoid intrusive questions. For example, use **age ranges** instead of exact ages, or omit unnecessary personal questions altogether.
- **Behave with Integrity**
  - **Accurate and Honest Reporting**: Record and report data **accurately** and **fully**, avoiding manipulation, falsification, or omission of data that does not support your case.
  - **Data Security and Ethical Use**: Ensure data is **securely stored** (e.g., password-protected, backed up) and inaccessible to unauthorized individuals. Consider the **potential harm** of research outcomes and disclose any **sponsors** to maintain transparency and trust.

- **Follow Appropriate Professional Codes of Conduct**
  - **Adherence to Codes**: Follow **professional codes of conduct** from relevant bodies (e.g., ACM, AIS, BCS, IEEE) to guide ethical decision-making in research.
  - **Cross-Disciplinary Guidance**: Utilize codes from **social science disciplines** (e.g., psychology, sociology) for research involving human participants, even if not a member of those professional bodies.

**Examples of professional codes of conduct relevant to IS and computing research**

| | |
|---|---|
| ACM | Association for Computing Machinery<br>www.acm.org |
| AoIR | Association of Internet Researchers<br>www.aoir.org |
| AIS | Association for Information Systems<br>www.aisnet.org |
| APA | American Psychological Association<br>www.apa.org |
| ASA | American Sociological Association<br>www.asanet.org |
| BCS | British Computer Society<br>www.bcs.org.uk |
| BPS | British Psychological Society<br>www.bps.org.uk |
| BSA | British Sociological Association<br>www.britsoc.co.uk |
| IEEE | Institute of Electrical and Electronics Engineers<br>www.ieee.org |

- **No Plagiarism**
  - **Avoid Stealing Credit**: Do not present someone else's work as your own, as it is considered **unethical** and akin to stealing.
  - **Proper Attribution**: Always provide **full credit** to the original author with clear references, enabling readers to locate the source material.
  - **Avoid Self-Plagiarism**: Do not submit a paper for publication that is very similar to one you have already published without proper disclosure.
  - **Transparency with Editors**: If submitting a revised conference paper to a journal, inform the editor about the earlier version and acknowledge it in a footnote.
  - **Single Submission Policy**: Avoid submitting the same manuscript to multiple journals simultaneously, as it wastes reviewers' time and is considered unethical.
  - **Ethical Writing Practices**: Ensure all work is original, properly cited, and transparently disclosed to maintain **integrity** and **credibility** in academic publishing.

- **Be an Ethical Reviewer**
  - **Timely Reviews**: Complete reviews **promptly** to avoid unnecessary delays in the publication process and respect the authors' time.
  - **Maintain Confidentiality**: Keep the content of the manuscript **confidential**, including ideas and data, and do not use it for personal gain before publication.
  - **Professional Feedback**: Provide **clear, constructive, and professional feedback**, avoiding illegible or overly critical comments. Highlight at least one positive aspect of the work.
  - **Helpful Suggestions**: Offer **useful recommendations**, such as relevant citations or improvements, to assist authors in enhancing their manuscript.
  - **Respect Authors' Efforts**: Approach the review process with **respect** for the authors' hard work and dedication, even if recommending rejection.
  - **Avoid Bias**: Ensure reviews are **objective** and free from personal biases, focusing on the quality and contribution of the research.

# Design and Creation Projects and Ethics

- **Unique Ethical Challenges**: Design and creation research in IS and computing involves creating **IT systems** or models, raising additional ethical concerns beyond data collection and reporting.
- **Misuse of Digital Data**: Digital data can be easily **accessed**, **copied**, or repurposed without consent, potentially violating privacy or intellectual property rights.
- **Privacy and Anonymity Risks**: IT systems can enable **unethical or unlawful acts** due to the anonymity they provide, making it harder to hold wrongdoers accountable.
- **Covert Data Gathering**: Technologies like **digital cameras**, **keystroke monitors**, **cookies**, and **RFID tags** allow covert observation, raising ethical concerns if used without **informed consent**.
- **Responsibility for System Use**: Researchers must ensure their designed systems are not used unethically by clients or sponsors, taking responsibility for the **ethical implications** of their work.
- **Awareness of Temptations**: Researchers must be vigilant about the **ethical risks** posed by IT systems and actively work to prevent misuse or harm.

# Data Privacy and Security in Research

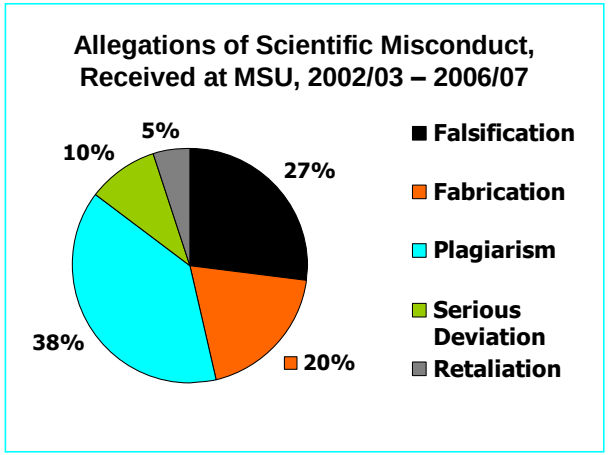
- **Data Privacy**: Ensure participants' **personal information** is handled with care, respecting their right to privacy and confidentiality throughout the research process.
- **Legal Compliance**: Adhere to **data protection laws** (e.g., GDPR, HIPAA) to safeguard sensitive data and avoid legal or ethical violations.
- **Secure Data Storage**: Use **encryption**, **password protection**, and secure servers to prevent unauthorized access or data breaches.
- **Informed Consent**: Clearly communicate to participants how their data will be **collected**, **used**, and **stored**, ensuring they provide informed consent.
- **Anonymization**: Remove or disguise **identifiable information** to protect participants' identities while maintaining the integrity of the research data.
- **Access Control**: Limit access to sensitive data to **authorized personnel only**, reducing the risk of misuse or accidental exposure.
- **Data Breach Protocols**: Establish clear procedures for responding to **data breaches**, including timely notification of affected participants and corrective actions.
- **Ethical Responsibility**: Researchers must prioritize **data privacy and security** to maintain trust, uphold ethical standards, and ensure the credibility of their work.

# What is Plagiarism?

- Plagiarism means “the appropriation of another person’s ideas, processes, results, or words without giving appropriate credit”
  - Source: MSU’s *Procedures Concerning Allegations of Misconduct in Research and Creative Activities* http://rio.msu.edu/June_2009_Procedures.pdf
- Appropriation means using or taking something that is not yours
  - Plagiarism is stealing by using another person’s words or ideas
  - Plagiarism is academically dishonest because students, scholars, and faculty members are expected to do their own work
- The terms ideas, processes, results, or words refer to another person’s intellectual property
- Plagiarizers use or take intellectual property
- Other thieves use or take physical property such as money, computers, things, etc.



- Plagiarism means taking or using any of the following intellectual property without permission or giving credit:
  - Words … copying more than 4-6 consecutive words, rearranging phrases, or paraphrasing extensively
  - Ideas … using original information learned from conference presentations, confidential reviews, etc.
  - Processes … adopting or using research methods described by another investigator, especially when the research method is not common knowledge
  - Results … using or reporting data, figures, or tables that represent another investigator's research results
- Giving appropriate credit refers to:
  - Providing the name of the original author, artist, researcher, or scholar
  - Providing sufficient publication data that another person can find the original source – **use disciplinary standards**!
  - Using quotation symbols to indicate direct quotes



# Intellectual property

- Copyright
- Industrial Property
  - Trademarks
  - Patent
  - Industrial designs
  - Confidential information
  - Geographical Indications
- IP as intangible property
  - Tangible property: Land, houses, car
  - Intangible property
    - intellectual property: Intangible wealth, easily appropriated and reproduced, once created the marginal cost of reproduction is negligible

- The role of IP as intangible property
  - economic rights of creators
  - commercial exploitation of owner of IP
  - capital expenditure
  - transfer of technology
  - cultural development
- Why IP protection is given
  - Capital expenditure for new products
  - R and D
  - Marketing and advertisement
  - No free loaders
  - Maintaining loyal followers
  - profit
- IP as a property
  - Can be sold
  - Can be bought
  - Can be lease or rent
  - Can pass under a will
  - Can be assigned

# Plagiarism and Intellectual Property

- **Intellectual Property (IP) Basics**:
  - IP refers to creations of the mind (e.g., inventions, literary works, designs) protected by **copyrights**, **patents**, or **trademarks**.
  - Under the **Berne Convention**, creators automatically own the copyright to their original work as soon as it is created, without needing to register it.
- **What is Plagiarism?**:
  - Plagiarism is the **theft of ideas or words** and the **fraudulent presentation** of someone else's work as your own.
  - It includes copying text, paraphrasing without credit, or reusing your own previous work without permission (**self-plagiarism**).
- **Why Avoid Plagiarism?**:
  - **Ethical Responsibility**: Plagiarism undermines trust, originality, and intellectual growth.
  - **Academic and Professional Consequences**: It can lead to failing grades, expulsion, job loss, lawsuits, or public humiliation.
  - **Societal Impact**: Plagiarism stifles innovation and economic growth by devaluing original ideas.
- **Types of Plagiarism**:
  - **Intentional Plagiarism**: Deliberately copying or buying work and submitting it as your own.
  - **Unintentional Plagiarism**: Failing to properly cite sources due to confusion, carelessness, or lack of understanding.
  - Examples include **copy-pasting**, **mash-ups** (mixing copied material), and **recycling** (reusing your own work without permission).
- **How to Avoid Plagiarism**:
  - **Cite Sources**: Always credit the original author for ideas, words, or data.
  - **Use Quotation Marks**: Clearly indicate direct quotes.
  - **Paraphrase Properly**: Rewrite ideas in your own words and still cite the source.
  - **Track Sources**: Keep detailed notes of all references used during research.
- **Common Knowledge vs. Citations**:
  - **Common Knowledge**: Facts widely known (e.g., "The Civil War began in 1861") do not require citations.
  - **Contested or Complex Ideas**: Always cite sources for opinions, interpretations, or controversial information.

# Algorithmic Bias and Fairness

- What is NOT bias in machine learning?
  - It is **not necessarily malicious**.
    - Bias can occur even when everyone, from the data collectors to the engineers to the medical professionals, have the best intentions.
  - It is **not one and done**.
    - Just because an algorithm has no bias now does not mean it has no potential later.
  - It is **not new.**
    - Researchers have raised concerns over the last 50 years.
- What IS bias in machine learning?
  - It is defined many ways, for example **disparate treatment or impact of algorithm**. See also, fairness or discrimination.
  - It is the **culmination of a flawed system**.
    - Sources including bias in the data collection, bias in the algorithmic process, and bias in the deployment.
  - It is the **vigilance** of how technology can amplify or create bias.

# Algorithmic Bias and Fairness

- How do we define “bias”?
  - Fairness through unawareness
  - Group fairness
  - Calibration
  - Error rate balance
  - Representational fairness
  - Counterfactual fairness
  - Individual fairness

The bias arises because the algorithm predicts health care costs rather than illness … despite health care cost appearing to be an effective proxy for health

**Dissecting racial bias in an algorithm used to manage the health of populations**

ZIAD OBERMEYER, BRIAN POWERS, CHRISTINE VOGELI, AND SENDHIL MULLAINATHAN Authors Info & Affiliations

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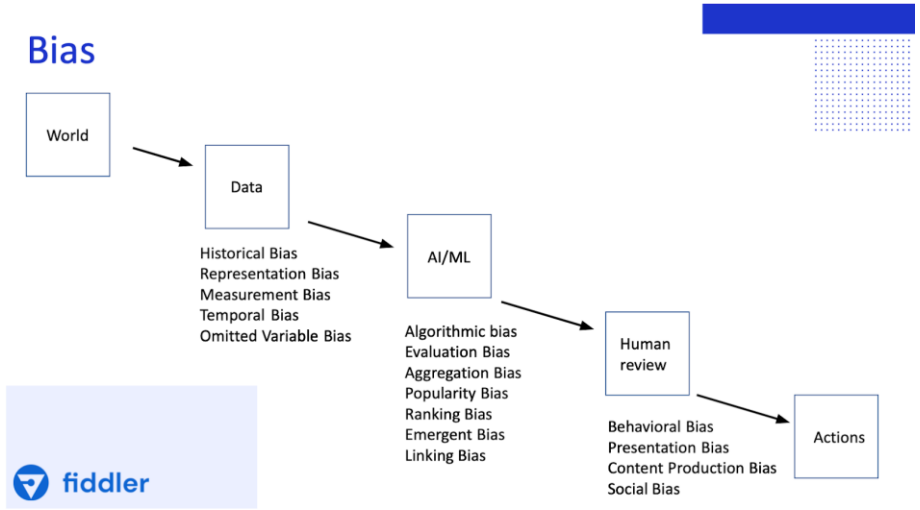
**Racial bias in health algorithms**

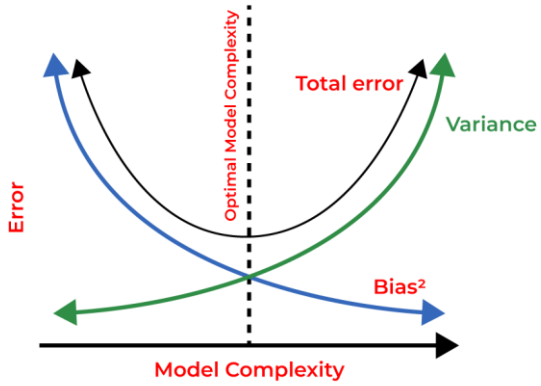
The U.S. health care system uses commercial algorithms to guide health decisions. Obermeyer *et al.* find evidence of racial bias in one widely used algorithm, such that Black patients assigned the same level of risk by the algorithm are sicker than White patients (see the Perspective by Benjamin). The authors estimated that this racial bias reduces the number of Black patients identified for extra care by more than half. Bias occurs because the algorithm uses health costs as a proxy for health needs. Less money is spent on Black patients who have the same level of need, and the algorithm thus falsely concludes that Black patients are healthier than equally sick White patients. Reformulating the algorithm so that it no longer uses costs as a proxy for needs eliminates the racial bias in predicting who needs extra care.

*Science*, this issue p. 447; see also p. 421

# Algorithmic Bias and Fairness

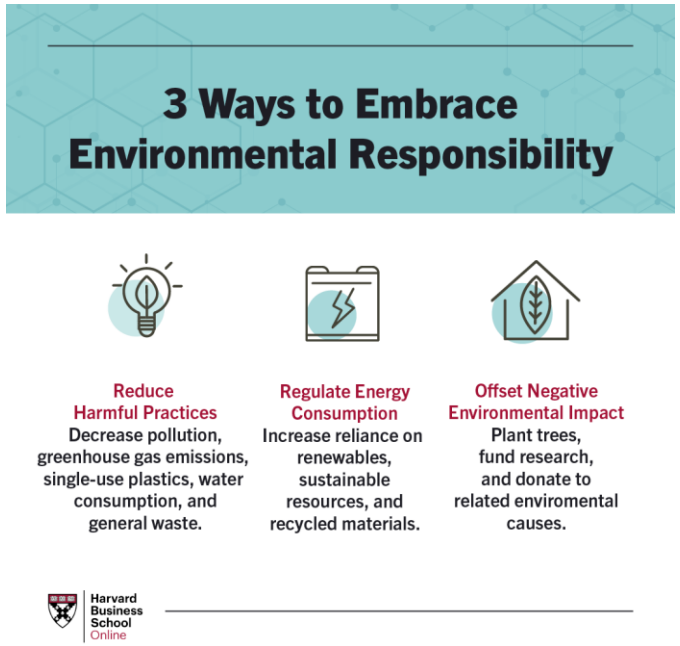
- Algorithmic bias occurs when an algorithm produces **unfair or discriminatory outcomes**, often due to biased data, design flaws, or unintended consequences in the algorithm's logic.
- **Causes of Algorithmic Bias**:
  - **Biased Training Data**: If the data used to train an algorithm reflects historical biases or inequalities, the algorithm will replicate or amplify these biases.
  - **Flawed Design**: Poorly designed algorithms may overlook important variables or overemphasize irrelevant ones.
  - **Lack of Diversity**: Teams developing algorithms may lack diversity, leading to blind spots in addressing fairness.
- **Examples of Algorithmic Bias**:
  - **Hiring Algorithms**: Favoring certain demographics over others based on biased historical hiring data.
  - **Predictive Policing**: Targeting specific communities disproportionately due to biased crime data.
  - **Facial Recognition**: Higher error rates for people of color or women due to underrepresentation in training datasets.
- **Consequences of Algorithmic Bias**:
  - **Reinforces Inequality**: Perpetuates existing social, economic, and racial disparities.
  - **Loss of Trust**: Erodes public trust in technology and institutions using biased algorithms.
  - **Legal and Ethical Risks**: Organizations may face lawsuits, reputational damage, or regulatory penalties.
- **Ensuring Algorithmic Fairness**:
  - **Diverse and Representative Data**: Use datasets that accurately reflect the diversity of the population.
  - **Bias Detection Tools**: Implement tools to identify and mitigate bias during algorithm development.
  - **Transparency and Accountability**: Make algorithms explainable and allow for external audits.
  - **Inclusive Design Teams**: Ensure diverse teams are involved in developing and testing algorithms.
- **Ethical Responsibility**:
  - Developers and organizations must prioritize **fairness**, **transparency**, and **accountability** to ensure algorithms do not harm individuals or groups.
  - Regularly review and update algorithms to address emerging biases and ensure compliance with ethical standards.

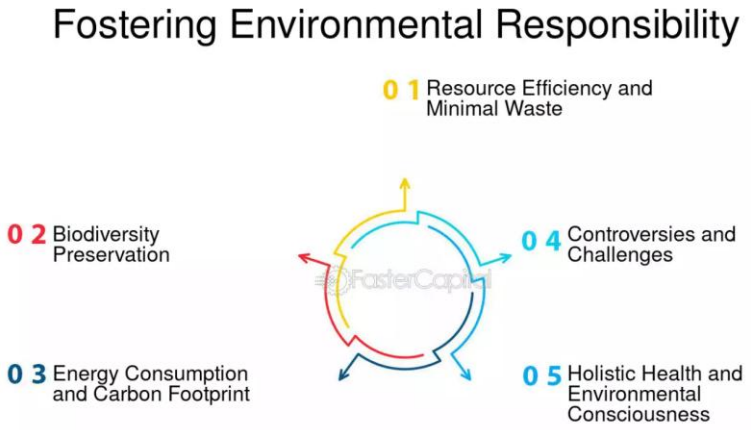


# Environmental Responsibility

- Environmental responsibility refers to the ethical obligation of individuals, organizations, and governments to protect and preserve the environment. This includes minimizing negative impacts on ecosystems, reducing pollution, and promoting sustainable practices.
- **Corporate Environmental Responsibility**:
  Businesses are increasingly adopting environmentally responsible practices, such as reducing carbon emissions, minimizing waste, and using renewable energy sources. This not only benefits the environment but also enhances their reputation and meets consumer expectations.
- **Conservation and Biodiversity**:
  A key aspect of environmental responsibility is the conservation of natural resources and biodiversity. This involves protecting endangered species, restoring habitats, and ensuring the sustainable use of resources like water, forests, and minerals.
- **Regulatory Compliance and Standards**:
  Governments and international organizations have established regulations and standards to enforce environmental responsibility. Compliance with laws such as the Clean Air Act, Kyoto Protocol, and Paris Agreement is essential for reducing environmental harm.
- **Individual Actions and Awareness**:
  Individuals play a crucial role in environmental responsibility by adopting eco-friendly habits, such as recycling, reducing energy consumption, and supporting sustainable products. Public awareness and education are vital for driving collective action.
- **Long-Term Sustainability**:
  Environmental responsibility is closely tied to the concept of sustainability, which focuses on meeting present needs without compromising the ability of future generations to meet theirs. This requires a balance between economic growth, social well-being, and environmental protection.



# Conflict of Interest

- A conflict of interest occurs when personal, financial, or professional interests interfere, or appear to interfere, with the integrity, objectivity, or impartiality of research or decision-making.
- **Types of Conflicts**:
  - Conflicts can be **individual** (e.g., financial stakes, interpersonal relationships),
  - **institutional** (e.g., financial or reputational interests), or
  - **REB-related** (e.g., reviewing one's own research).
- **Impact on Research Integrity**: Conflicts of interest can undermine the credibility of research, erode public trust, and compromise participant welfare. They may also violate ethical principles such as Respect for Persons and Justice.
- **Disclosure Requirement**: Researchers, institutions, and REB members must disclose real, potential, or perceived conflicts of interest to ensure transparency and accountability.
- **Institutional Responsibilities**: Institutions must develop and implement policies to identify, eliminate, minimize, or manage conflicts of interest. These policies should be publicly accessible and clearly define roles and responsibilities.
- **Dual Roles**: Researchers acting in dual roles (e.g., as both a researcher and a therapist or teacher) must disclose potential conflicts to participants to avoid power imbalances or coercion.

# Conflict of Interest in Research

- **Definition**:
  - A conflict of interest occurs when financial or personal factors impair or appear to impair a researcher's professional judgment in conducting or reporting research.
  - It is defined by the circumstances, not the individual's actions or character.
- **Ethical Dilemma**: Unmanaged conflicts cast doubt on research impartiality and decision-making.
- **Examples**:
  - Conducting evaluative research for a firm in which the researcher has a financial stake.
  - Accepting funding from a company with a vested interest in the research outcomes.
  - Negotiating with a company where the researcher or their relatives have a monetary interest.
- **Impact**:
  - Conflicts of interest can damage public trust in researchers and the scientific community.
  - Even unintentional wrongdoing can harm a researcher's reputation, funding, and career.
  - Perceptions of dishonesty can be as damaging as actual misconduct.
- **Types**:
  - **Financial**: Monetary ties that may influence research outcomes (e.g., funding, stock ownership).
  - **Personal**: Relationships or beliefs that affect research (e.g., family ties, ideological biases).
  - **Professional**: Situations that provide a professional advantage (e.g., reviewing a competitor's work).
- **Prevention**:
  - Disclose financial support, personal activities, and institutional ties.



# Intellectual Property Again

- **Patent**
- **Copyright**
- Trademarks
- Trade Names
- Trade Dress
- Federal rights
- International protections
- Common law rights

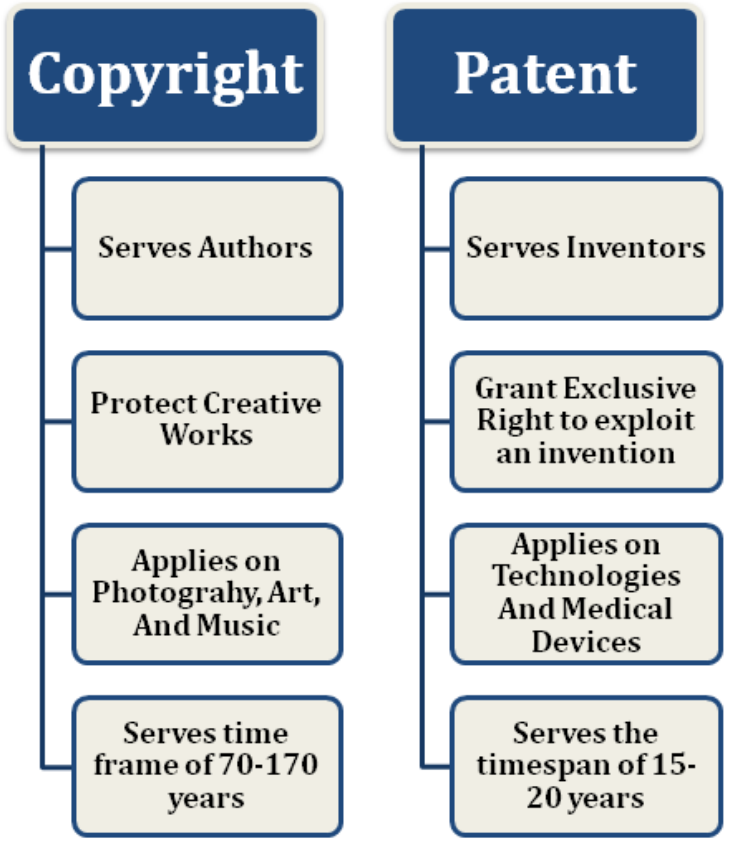
- **Patent**
  - A legal Monopoly granted by government.
  - Allows the inventor an exclusive right to make, use, and sell an invention for twenty (20) years.
  - Patent applicant must demonstrate to the patent office that an invention, discovery, or design is non-obvious, novel, and useful. It cannot be obvious considering current technology.
- Utility or function patent protects “the way an article is used or works.” Must be reduced to tangible form.
- Plant patent protects new ”asexually reproduced distinct variety of plant, other than a tuber propagated plant or a plant found in an uncultivated state.”
- **Design patents protects the way an article looks.**
  - **Lesser term protection - 14 years.**
  - **“consists of the visual ornamental characteristics embodied in, or applied to, an article of manufacture.”**

# Patent

- To obtain a patent, the new invention must be:
  - Novel – not known or used in this country and not published anywhere.
  - Nonobvious – cannot be an obvious way to do something.
  - Useful – must have some application, even if not commercially practical.
- Priority Between Two Inventors – generally, the person first to invent and use the product is given the patent, even over an earlier filer.
- Prior Sale –must apply for a patent within one year of selling the product.
- Provisional Patent Application – a shorter, cheaper way to file for a patent temporarily, to determine if the invention is commercially practical.
- **Patent office grant of the patent is subject to judicial review in a patent infringement case.**

# Copyright

- An intangible right given to the author or originator of certain literary or artistic productions.
- Works created after January 1, 1978, are automatically given statutory copyright protection for the life of the author plus seventy years.
- Copyrights owned by publishing houses expire ninety-five years after publication or a one hundred-twenty-years from the date of creation, whichever is first.
- If multiple authors, expiration is seventy years after the death of the last surviving author
- Facts are not copyrightable
- The employer is the owner of works for hire-work created within the scope of employment unless otherwise agreed
- The **Berne convention** provides international protection for copyright for citizens of the 96 signatory countries http://www.cerebalaw.com/Berne.htm
- Exists automatically upon creation of an original work but
- Placement of copyright symbol © gives notice and to
- judicially enforce copyright requires Registration with the US Copyright Office



# ACM Code of Ethics

| | |
|---|---|
| **1** | **General moral imperatives** |
| 1.1 | Contribute to society and human well-being |
| 1.2 | Avoid harm to others |
| 1.3 | Be honest and trustworthy |
| 1.4 | Be fair and take action not to discriminate |
| 1.5 | Honor property rights including copyrights and patents |
| 1.6 | Give proper credit for intellectual property |
| 1.7 | Respect the privacy of others |
| 1.8 | Honor confidentiality |
| **2** | **More specific professional responsibilities** |
| 2.1 | Strive to achieve the highest quality, effectiveness and dignity in both the process and products of professional work |
| 2.2 | Acquire and maintain professional competence |
| 2.3 | Know and respect existing laws pertaining to professional work |
| 2.4 | Accept and provide appropriate professional review |
| 2.5 | Give comprehensive and thorough evaluations of computer systems |
| | and their impacts, including analysis of possible risks |
| 2.6 | Honor contracts, agreements and assigned responsibilities |
| 2.7 | Improve public understanding of computing and its consequences |
| 2.8 | Access computing and communication resources only when authorised to do so |

VISIT:
https://www.acm.org/code-of-ethics

# IEEE Code of Ethics

We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

I. **To uphold the highest standards of integrity, responsible behavior, and ethical conduct in professional activities.**

1. to hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment;

2. to improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, including intelligent systems;

3. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;

4. to avoid unlawful conduct in professional activities, and to reject bribery in all its forms;

5. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, to be honest and realistic in stating claims or estimates based on available data, and to credit properly the contributions of others;

6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;

https://www.ieee.org/about/corporate/governance/p7-8.html

II. **To treat all persons fairly and with respect, to not engage in harassment or discrimination, and to avoid injuring others.**

7. to treat all persons fairly and with respect, and to not engage in discrimination based on characteristics such as race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;

8. to not engage in harassment of any kind, including sexual harassment or bullying behavior;

9. to avoid injuring others, their property, reputation, or employment by false or malicious actions, rumors or any other verbal or physical abuses;

III**. To strive to ensure this code is upheld by colleagues and co-workers.**

10. to support colleagues and co-workers in following this code of ethics, to strive to ensure the code is upheld, and to not retaliate against individuals reporting a violation.

Adopted by the IEEE Board of Directors and incorporating revisions through June 2020.